DISTANCE PERCEPTION USING STEREOGRAPHIC METHOD

Project work submitted in partial fulfilment
of the requirements for the award of
Degree of Bachelor of Engineering in
Computer Science & Engineering
of Madurai Kamaraj University.

Submitted by

A. SARAVANAN (94C42)

Under the guidance of

MISS. G.ANDAL JEYALAKSHMI B.E., M.S.,



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THIAGARAJAR COLLEGE OF ENGINEERING
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BONAFIDE CERTIFICATE

Certified that this is a bonafide record of the Project work done by

A. SARAVANAN (94C42)

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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We also thank all our friends for extending their moral support during this project work.

Dedicated to our beloved Parents



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SYNOPSIS

SYNOPSIS

1. SYNOPSIS

Our project is to make the machine to perceive the exact location of a point in the real - world.

This is the replication of visual capabilities of human being on a computer.

This is possible by first taking two photographs of a situation in the real world, from two different views, for which the distances between those places are known.

The images were stored as **Bitmap files** and analysed for the occurance of a specific point. After finding that point the system will adopt a mathematical model and calculate the distance of that point from the camera. The mathematical model connects the image points and the real point in the real world which results in a geometrical figure by which the system is able to distiinguish a point.

This shows us that things related with sense - data can be brought to machines by using simple but beautiful mathematial formalism.

SYSTEM STUDY

2. SYSTEM STUDY

2.1. Historical Introduction

In those days Before Christ people were very fond of knowing how far those mystical celestial bodies were from earth.

This is done first by Greeks using Stereography method. They observed the relation between, the disparity of a thing when viewed from left and right eye, and its distance.

They used the same technique for finding the distance of space objects by observing from two places. But they must make measurements like angles during observation, by which they later calculate the distance.

Later in Seventeenth Century Philosopher Leibnitz introduced 'monadology' which describe how brains manipulate monads. Monads were mind - equivalent of world things.

Then in Twentieth Century Mathematician and Philosopher 'Bertrand Russell'extented 'monadology' and explained how brains analyse monads and how it convolve monads to know the distance of that thing. But monads have no common element. So a problem arises and he sees that monadology is not sufficient.

So Russell give more importance to sense - data rather than monads. So sensiblia is the key for understanding the distance perception. sensiblia will have common elements. So Brain will identify those common elements and perceive the distance.

This give us an idea for the replication of the above to machines.

2.2. Introduction to our system

The computer has no sensiblia receiver for that we've used cameras which is eye - equivalent.

So using cameras, for eyes, and algorithm for computer which is equivalnt to thought process in brain, and finally using the mathematical model we acheive machine vision. But in a very small scale because of limitations of present systems.

The Method

We take two - photographs of the real world from two places. We do this with a known distance between the centre of lenses of two cameras Here after the distance between them implicitly means the distance between the centre of lenses of two cameras.

The seperation of cameras is in horizontal direction only.

We store the Photos in Bitmap files. We then transfer information of photos to two arrays.

Now literally these arrays are machine - equivalent of

sensiblia.

After that we convole the two arrays for a point. This is done by searching a point in both arrays. Now the position of that point in photo can be described by cartesian coordinates by taking the centre as origin.

With this co-ordinates a simple geometrical structure, can be created which contain all the information we need.

Thus the system, eventhough the main operations were done manually, perceived the distance literally.

2.3. About bitmap files.

Two photos were scanned by a scanner and stored as Bitmap files.

Bitmap files will have mapped pixel information for a point in the scanned photo. Bitmap for file is an uncompressed file. This file is literally sensiblia so we have to analyse this. For that we have to access it and obtain what we need.

It has three header structures. The first one has a variable which is used to identify whether this is a BMP file or not. It also has a variable which tell us where the pixel information is located in this file. Detailed descriptions of all variables in these three headers is out of scope.

Next header has many variables. But we are interested in biwidth, biheight, bixpelspermeter, biypelspermeter, and bibitcount, Bibitcount indicate how many bits were used to represent a particualr color. **Biwidth, biheight** tell us the width and height of the photo in no of pixels. The remaining two variables specify no of pixels per meter in x direction and y direction.

The third header structure is for specifying what proportion of Red, Green and Blue has to be there for a specific color. One structure is needed for one color. So the structure is repeated as many times as the no of colors used in this file. No of colours used can be known from Second header variable binoofolors.

So this repeated structure forms a color - table. The data area when accessed will return a pointer which point to any colour in this color - table.

All data can be accessed using the pointer available. So all information is tranfered to a two dimensional arrays. We transfer also the other photos information to another array. Now we need to process these arrays and obtain what we need.

How this done is explained in subsequent chapters.

MATHEMATICAL MODEL

MATHEMATICAL MODEL

3. MATHEMATICAL MODEL

3.1. Introduction :

Cameras usually capture the object upside - down and rightside - left same as eyes, which all we know.

In this chapter whatever we discuss is based on negatives. But in subsequent chapters it may either be **negative or photo**. Because we store only photos in computer which is the reflection of an object.

Now refering to fig 3.1 we see that, **point** p was captured of by cameras and that point's image can be located by two sets of co-ordinates xl, yl and xr, yr. Here l and r stands for left and right negatives.

In the diagram and all following discussion we follow **the convention that we are facing the camera from real world**.

So for a person looking the real world behind camera, the left side for you is right side for him.

3.2. Calculating x and z

To perceive the location of any point we need three co-ordinates x y and z. We came to know these three by the following mathematical model.

Fig 3.2 is the top view of fig 3.1

In that points A and B the centres of cameras left and right.

There f is the distance between the camera lens centre and negative's centre.

And the known distance is denoted by d

With point's image co-ordinates following calculation were made.

$$\theta_1 = \tan^{-1}\left|\frac{f}{xl}\right| \qquad (3.1)$$

$$\theta_2 = \tan^{-1}\left|\frac{f}{xr}\right| \qquad (3.2)$$

We know that

$$\alpha = \theta_1 \qquad (3.3)$$

$$\beta = \theta_1 \qquad (3.4)$$

$$\gamma = 180 - (\alpha + \beta) \qquad (3.5)$$

For any triangle like Δ ABC here, the equality (3.6) is true.

$$\frac{\sin\alpha}{a} = \frac{\sin\beta}{b} = \frac{\sin\gamma}{d} \qquad (3.6)$$

a,b and d were sides opposite to angle α, β and γ.

For the proof of (3.6) See Appendix B and C.

We know angles α,β and γ and side d so other two sides were calculated as

$$a = \frac{\sin\alpha}{\sin\gamma} \times d \qquad (3.7)$$

$$a = \frac{\sin \beta}{\sin \gamma} \times d \qquad (3.8)$$

The real coordinaties x,y, z were calculated as below. Here we conventionally take the centre of lens as the origin.

As told already the cameras were seperated in x - direction. So two real co-ordinate set can be obtained but on both, y and z will be the same but x. So two x were calculated

$$Lx = b.\cos \alpha \qquad (3.9)$$

$$Rx = a.\cos \beta \qquad (3.10)$$

z can be calculated either as

$$z = \sqrt{b^2 - (Lx)^2} \quad \text{or as} \qquad (3.11)$$

$$z = \sqrt{a^2 - (Rx)^2} \qquad (3.12)$$

Theoritically both z values must be same. But due to manual errors incurred during shooting picture it may vary with negligible amount.

3.3. Calculating Y.

Fig 3.3 is the sideview of Fig 3.1 yl and yr were two image coordinates. With the help of one of these real y can be calculated. As said above the calculation by taking one y won't match with by taking other y. This is because of manual error as said above.

Now

$$\theta_3 = \tan^{-1}\left|\frac{yl}{f}\right| \qquad (3.13)$$

We know all three co-ordinates of that point now with this we can do the task of locating that point.

As said in intoduction all measurement were calculated only with images, when the obervation is over.

We can now know the angle at which the point is and its straight line distance from lens. So s - d - l (for left) and s - d - r (for right) were

$$s - d - l = \sqrt{(Lx)^2 + Y^2 + Z^2} \qquad (3.14)$$

$$s - d - r = \sqrt{(Rx)^2 + Y^2 + Z^2} \qquad (3.15)$$

The given equations and all that is theoretical how this is computerised is explained in subsequent chapters.

FIG 3.1 Representation of point falling on two negatives

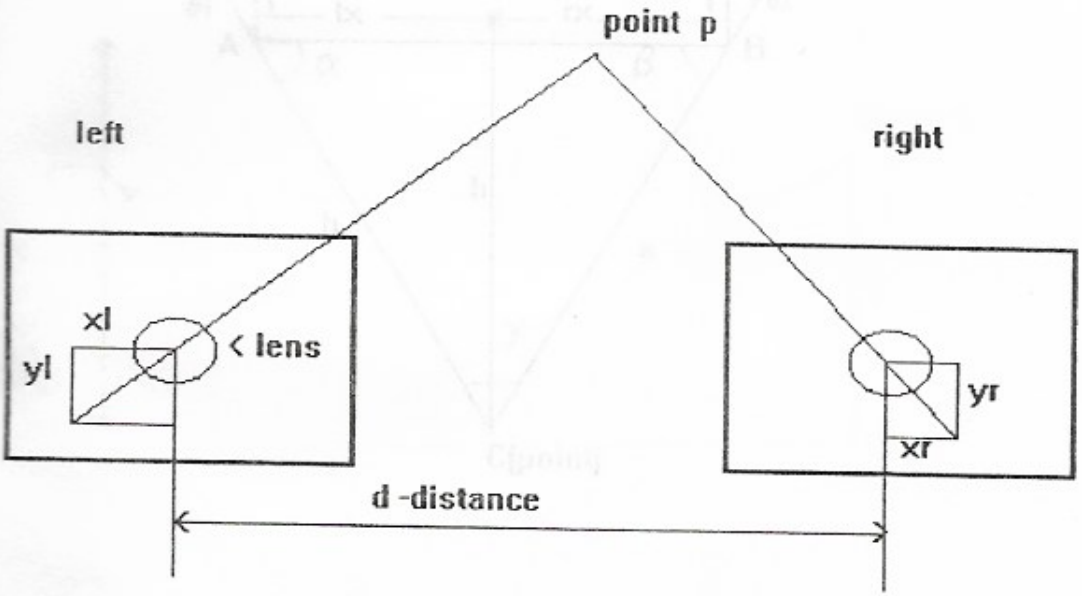


FIG 3.2 Top view of fig 3.1.

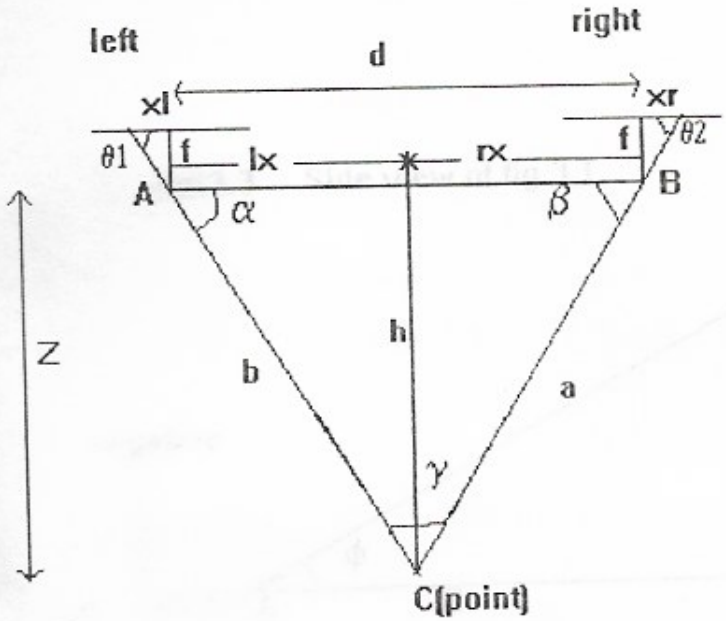
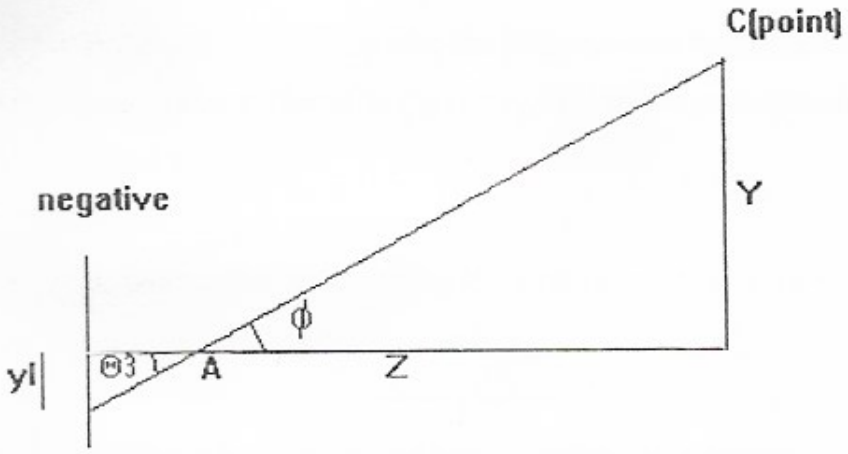


FIG3.3 Side view of fig 3.1.



SYSTEM DESIGN
& FLOW CHART

SYSTEM DESIGN & FLOW CHART

4. SYSTEM DESIGN & FLOW CHART

The photos were scanned by a black and white scanner and the files were stored as .BMP file. The steps followed in the processing of the images are.

1. Acces the .BMP file header structures to know the characteristics of the photo and information of all pixels.

2. An array is created with the size same as that of .BMP file and then all pixel information is transfered to that array. The above two operation is done twice for left and right photos.

3. Search the left array for the occurance of a point of object for which we are going to determine the distance.

4. If found, do the same for right array, else print error message as no - valuable point is there and exit.

5. If a matching point is found in right array also do the next step, else print error message as photos may be incorrect.

6. Obtain the image coordinates for array size.

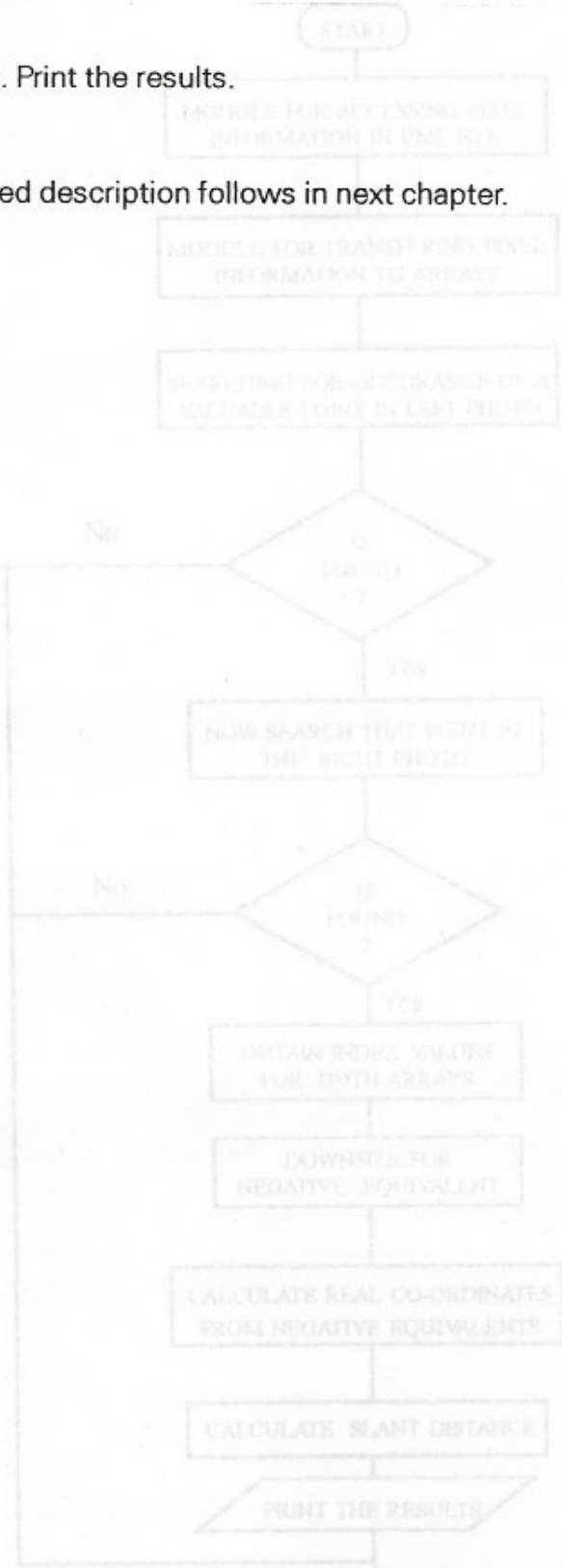
7. Convert that for negatiue size for calculation.

8. Calculate 3 - coordinates with xl, yl and xr, yr.

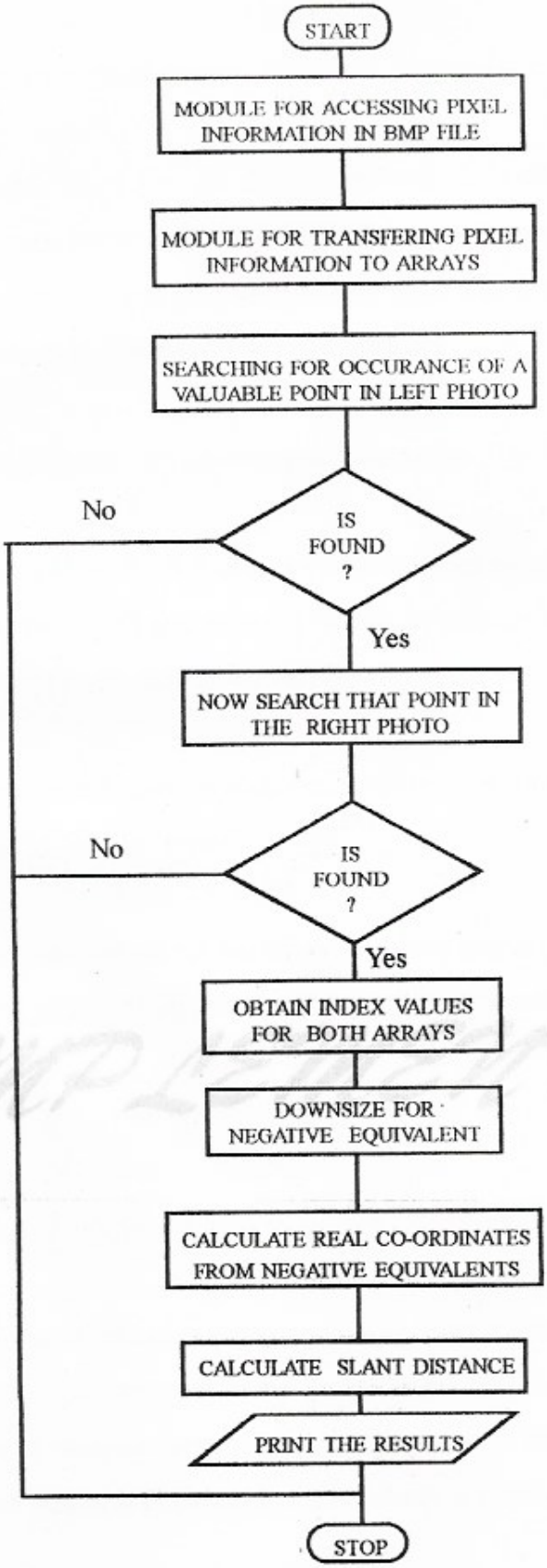
9. Now the computer came to know the position of that point 1.

10. Print the results.

A detailed description follows in next chapter.



FLOW CHART


START
MODULE FOR ACCESSING PIXEL INFORMATION IN BMP FILE
MODULE FOR TRANSFERING PIXEL INFORMATION TO ARRAYS
SEARCHING FOR OCCURANCE OF A VALUABLE POINT IN LEFT PHOTO
IS FOUND ?
No
Yes
NOW SEARCH THAT POINT IN THE RIGHT PHOTO
IS FOUND ?
No
Yes
OBTAIN INDEX VALUES FOR BOTH ARRAYS
DOWNSIZE FOR NEGATIVE EQUIVALENT
CALCULATE REAL CO-ORDINATES FROM NEGATIVE EQUIVALENTS
CALCULATE SLANT DISTANCE
PRINT THE RESULTS
STOP

IMPLEMENTATION

5. IMPLEMENTATION

5.1. Searching a valuable points

The valuable point may be defined differently in different contexts. we have used black & white scanner. So discrimination and decision on minor variations will lead to uncertanity. So we take very simple case that a valuable point is the point having color which is different from background.

Background color can be assigned manually.

When we see in fig 5.1 we can easily visualise that a point in side LR has counter point only in RR and point in RL has counterpoint only in LL. This reduces our whole search to an optimized search.

If a point is not found on both pair then we are go for a third alternative. That is searching in LL and RR sides.

The figure describes for negative and the array is photo-equivalent and 180 rotated image of negative. So the above optimized search on array is

a LL then RL

b RR then LR

c if not found by a or b then LR and RL.

Another optimization is when a valuable point is found in one array theoretically its counterpoint can also be found in the same row without searching any other row. But due to mannual error we give allowance so that our algorithm will search in 4 to 6 rows above and below the row where a point is found in other array.

5.1.2. DOWN SIZING AND CALCULATION

After a valuble point is found its coordinates were obtained. But this is enlarged picture and the model we have proposed is by taking negative co-ordinates. So we have to down size these coordinates to negative size.

Let Xb, Yb be the coordinate for size of .BMP file.

We know BMP file's width, height converting ratios (xpels permeter and ypelspermeter) with all these. We convert Xb - Yb, width, height from no - of pixels to meter or mm. Which is x - m, width - m, height - m.

So

$$Xm = \frac{1}{\text{Xpelspermeter}} \times b \text{ meter} \qquad (5.1)$$

$$Ym = \frac{1}{\text{Ypelspermeter}} \times b \text{ meter} \qquad (5.2)$$

$$Widthm = \frac{\text{width}}{\text{Xpelspermeter}} \text{ meter} \qquad (5.3)$$

$$Heightm = \frac{\text{height}}{\text{Ypelspermeter}} \text{ meter} \qquad (5.4)$$

From these metric converted variables we obtain xl, yl and xr, yr.

The camera we have used has negative size as 35mm x 24 mm. We call it w - n and h - n.

So

$$Xl = \frac{x - m}{width - m} \times \frac{h - n}{1000} \text{ meter} \qquad (5.5)$$

$$Yl = \frac{y - m}{height - m} \times \frac{w - n}{1000} \text{ meter} \qquad (5.6)$$

Similarly we can find xl,yl and xr,yr we can determine real co-ordinates x,y,z using formulae (3.1 to 3.12).

This with x,y,z our initial goal of seeing by computer using camera is achieved primitively.

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So

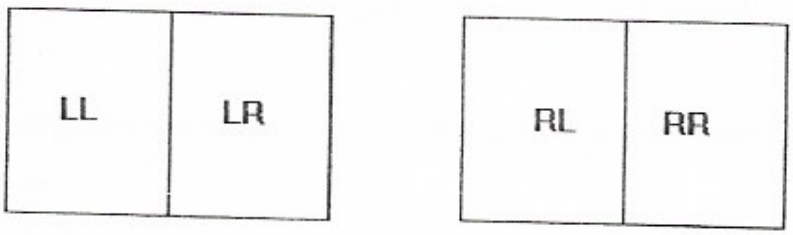
$$Xl = \frac{x - m}{width - m} \times \frac{h - n}{1000} \text{ meter} \qquad (5.5)$$

$$Yl = \frac{y - m}{height - m} \times \frac{w - n}{1000} \text{ meter} \qquad (5.6)$$

Similarly we can find xl,yl and xr,yr we can determine real co-ordinates x,y,z using formulae (3.1 to 3.12).

This with x,y,z our initial goal of seeing by computer using camera is achieved primitively.

FIG 5.1 Negatives.



APPLICATIONS

6. APPLICATIONS

It is possible to speculate future application which may evolve if it extend our concepts and put in more practise.

1. Vision is continuous so in future video information might be processed parallely. This can be done by a video processor which process the video information and interrupt the main procesor when a particular thing is recognized.

2. This is not a system for finding exact distance of anything. For that there are many engineering devices like SONAR. So this system will determine distance only approximately and have uncertain results which is a fuzzy nature. So this thing is for building intelligent machines rather than for precise machines. We humans were intelligent only because we have uncertainity, multivaluedness, Indeterminacy.

So our system when extended may initiate or impart fuzzy logic through visual sense - data.

3. Also another dynamical systems approach is Neural Networks. Which can be trained by visual data and can be made to recogize human faces or any other thing. So this will reduce the tedious work of spotting criminals or somebody, and be enough to do these things.

CONCLUSION

7. CONCLUSION

We have taken the photographs of the real world and tested our mathematical model, theoretically which gave precise results.

After the theoretical vertification we computerised that manual calculations. This is done by first scanning the photos by a black and white scanner and storing that as .BMP file. Then it is transferred to arrays. After that we have done successfully the crucial step of finding a valuable - point from a large array involving lakhs of elements.

Due to the present limitations of hardware we make the picture as simple as possible and involving only one figure.

Calculations were made on that point and the desired result which is the distance of that object from cameras were obtained.

We hope that in future we may extend our system to process the complex pictures which has thousands of distinguishable colours.

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8.BIBLIOGRAPHY

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Addison wesley Reading Massachusets 1992.

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Galgotia Publishers. New Delhi 1991.

s :\cse8 \95lc02 >jk

ENTER THE LEFT PHOTO : l.bmp

ENTER THE RIGHT PHOTO :r.bmp

* focal length of our camera = 35mm *

* following datas were based on left lens as reference *

RESULTS OBTAINED

The object`s location is in :

- left side and upper part.

It`s real co-ordinates are

x=2 meters y=0.5 meter z=6 meters

slant distance =7 meters

DO YOU WANT TO CONTINUE (Y/N) ?: N

s:\cse8 \95lc02 >

APPENDIXES

APPENDICES

APPENDICES

APPENDIX A

The geometry involved , if in Δ ABC β become an obtuse angle. A similar approach can also be given if angle α become obtuse.

From fig A.1

θ_1 = tan-1[f / xl] and θ_2 = tan-1[f / xl]

$\alpha = \theta_1$

$\beta = 180-\theta_2$

$\gamma = 180-(\alpha+\beta)$

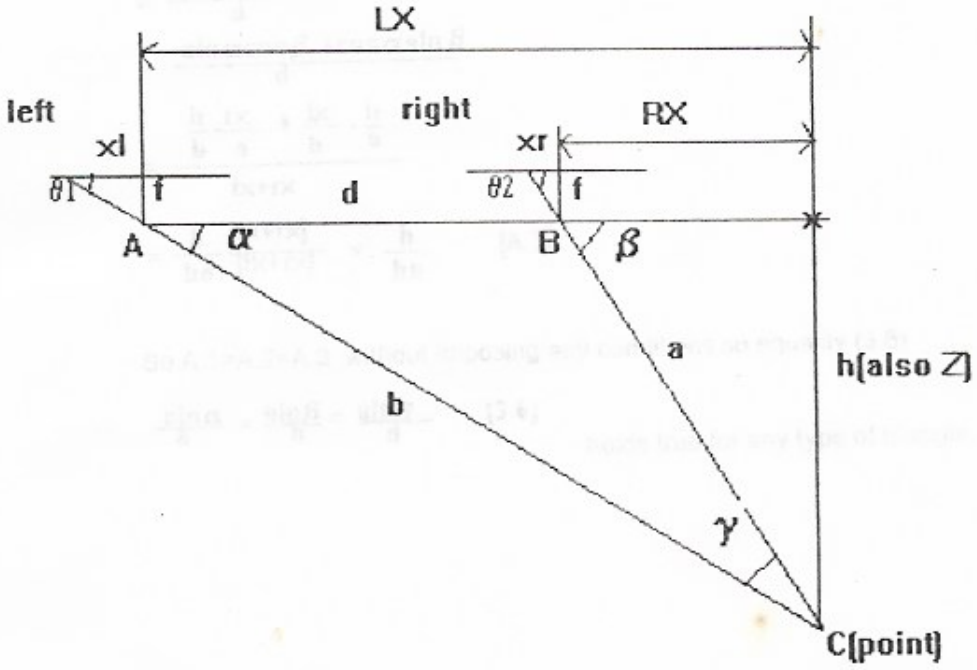
LX= b . cos(α)

RX=a . cos(180- β)

$Z=\sqrt{b^2 - (lx)^2}$ or

$Z=\sqrt{a^2 - (rx)^2}$

FIG A.1 Special case of fig 3.2 when β is an obtuse angle.



APPENDIX B

Proof for the equality (3.6) for all type of triangles

$$\frac{\sin\alpha}{a} = \frac{\frac{h}{b}}{a} = \frac{h}{ba} \qquad \text{(A.1)}$$

$$\frac{\sin\beta}{b} = \frac{\frac{h}{a}}{b} = \frac{h}{ba} \qquad \text{(A.2)}$$

$$\frac{\sin\gamma}{d} = \frac{\sin[180-(\alpha+\beta)]}{d}$$

$$= \frac{\sin[\alpha+\beta]}{d}$$

$$= \frac{\sin\alpha\cos\beta + \cos\alpha\sin\beta}{d}$$

$$= \frac{\frac{h}{b}\cdot\frac{rx}{a} + \frac{lx}{b}\cdot\frac{h}{a}}{lx+rx}$$

$$= \frac{h}{ba}\frac{(lx+rx)}{(lx+rx)} = \frac{h}{ba} \qquad \text{(A.3)}$$

So A.1=A.2=A.3 without imposing any conditions.so equality (3.6)

$$\frac{\sin\alpha}{a} = \frac{\sin\beta}{b} = \frac{\sin\gamma}{d} \qquad (3.6)$$

holds true for any type of triangle.

APPENDIX C

Proof for the equality (3.6) if in ΔABC angle β become an obtuse angle .The same treatment can be given for α if it become obtuse.
See fig A.1

$$\frac{\sin\alpha}{a} = \frac{\frac{h}{b}}{a} = \frac{h}{ba} \qquad \text{(A.1)}$$

$$\frac{\sin\beta}{b} = \frac{\frac{h}{a}}{b} = \frac{h}{ba} \qquad \text{(A.2)}$$

$$\gamma = 180-(\alpha+(180-\beta))$$
$$= 180-\alpha-180+\beta \quad = \beta-\alpha$$

$$\frac{\sin\gamma}{d} = \frac{\sin(\beta-\alpha)}{d}$$

$$= \frac{\sin\beta\cos\alpha - \cos\beta\sin\alpha}{d}$$

$$= \frac{\frac{h}{a}\cdot\frac{lx}{b} - \frac{h}{b}\cdot\frac{rx}{a}}{(lx-rx)}$$

$$= \frac{h}{ba}\cdot\frac{(lx-rx)}{(lx-rx)} = \frac{h}{ba} \qquad \text{(A.3)}$$

So the equality holds true.